

R-2R DAC based 4-bit ADC

Team Schmith

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Team members:

Srijith Reddy Pakala
Abhiram Avadhanam
Yuchao Wang

Team background

- Experience

Srijith: B.Tech in EE from IIT Hyderabad (currently Analog Char Engineer at Texas Instruments)

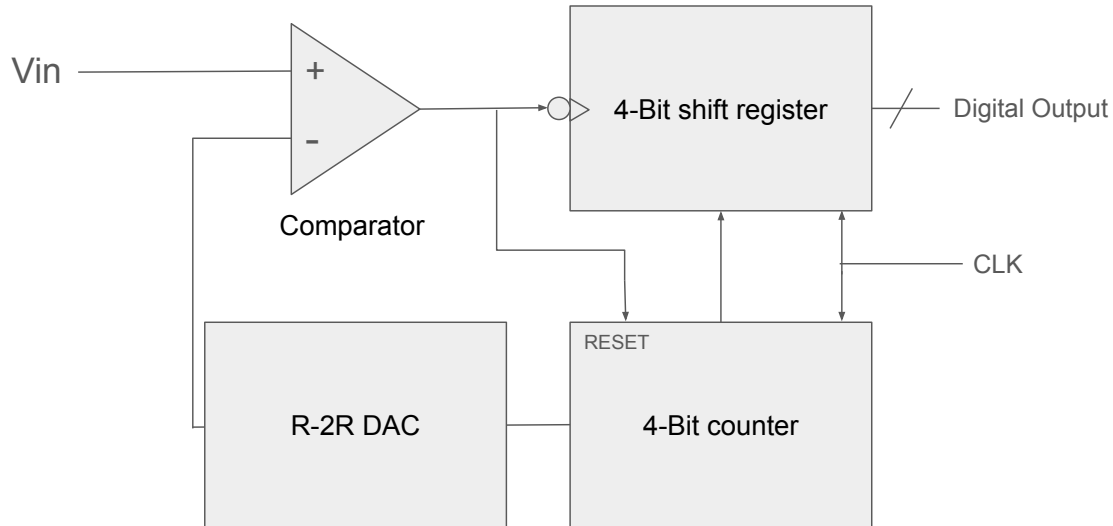
Abhiram : B.Tech in EE from Visvesvaraya College of Engineering (currently DV Engineer at Texas Instruments)

Yuchao: Graduate student in microelectronics at Chemnitz University of Technology, germany

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Project information

- Goal:
 - Build a 4-bit ADC using counter and R-2R DAC (Target sampling rate = 1MSPS)
- Design - High Level Proposal:
 - 2 stage comparator (reconfigurable)
 - DAC implementation based on resistor ladder that can be designed internally as well as externally
 - 4-bit counter and shift register (Non configurable)
 - The chip can utilise internal DAC ladder or external by setting a mode respectively

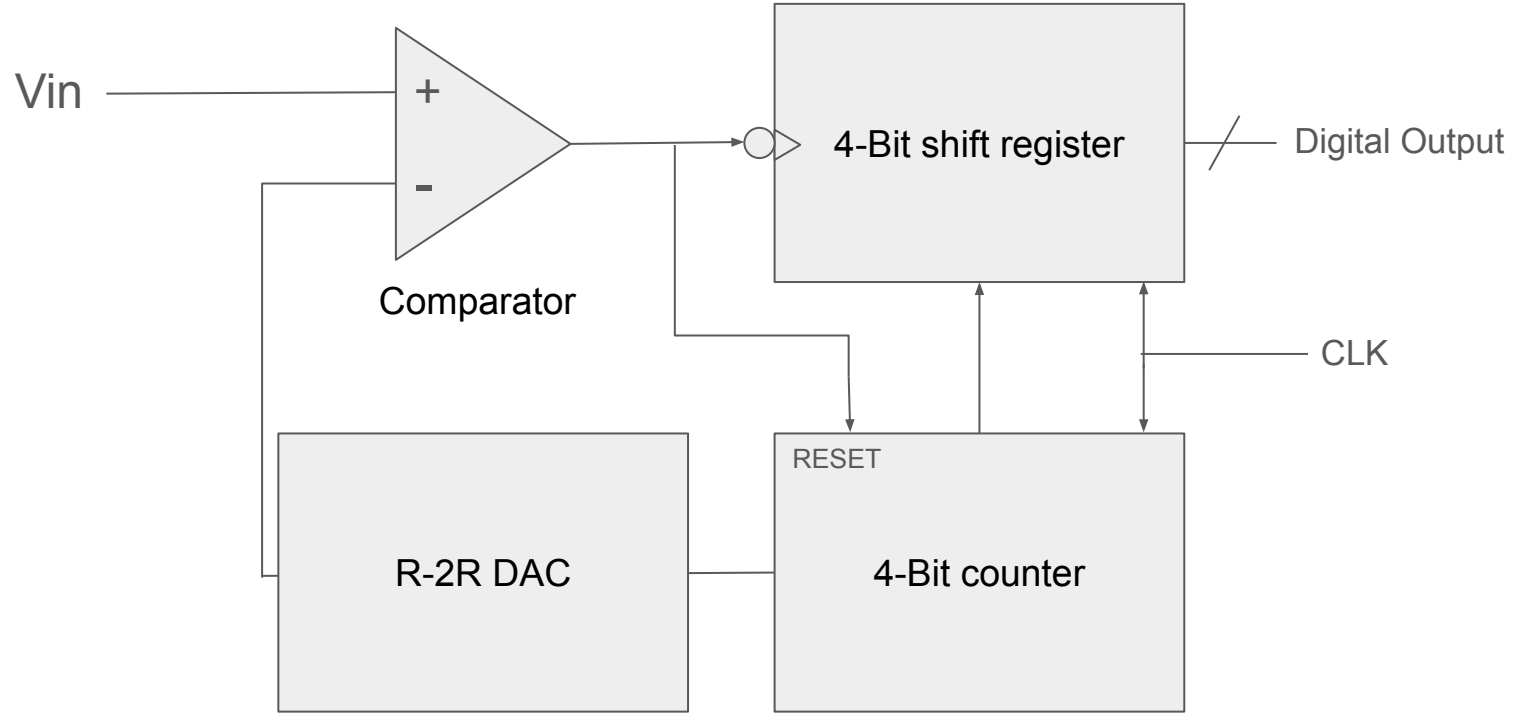


- Applications:
 - This circuit is designed to quantize analog signals and can be used in any frontends
 - Some of the functionality of the original MOSBius is also retained and can support those experiments as well
- References:
- GitHub repository: <https://github.com/SRIJITH01/Team-schmith/tree/main>
- Proposal detail: https://docs.google.com/presentation/d/18ij_HGlgQ7wV6nb4jpw9QSqh5o-CbtIRGRM82pGzgXk/edit?slide=id.g36dcc7f3034_1_215#slide=id.g36dcc7f3034_1_215
- Video link : [Link](#)

Questions, Suggestions, Doubts?

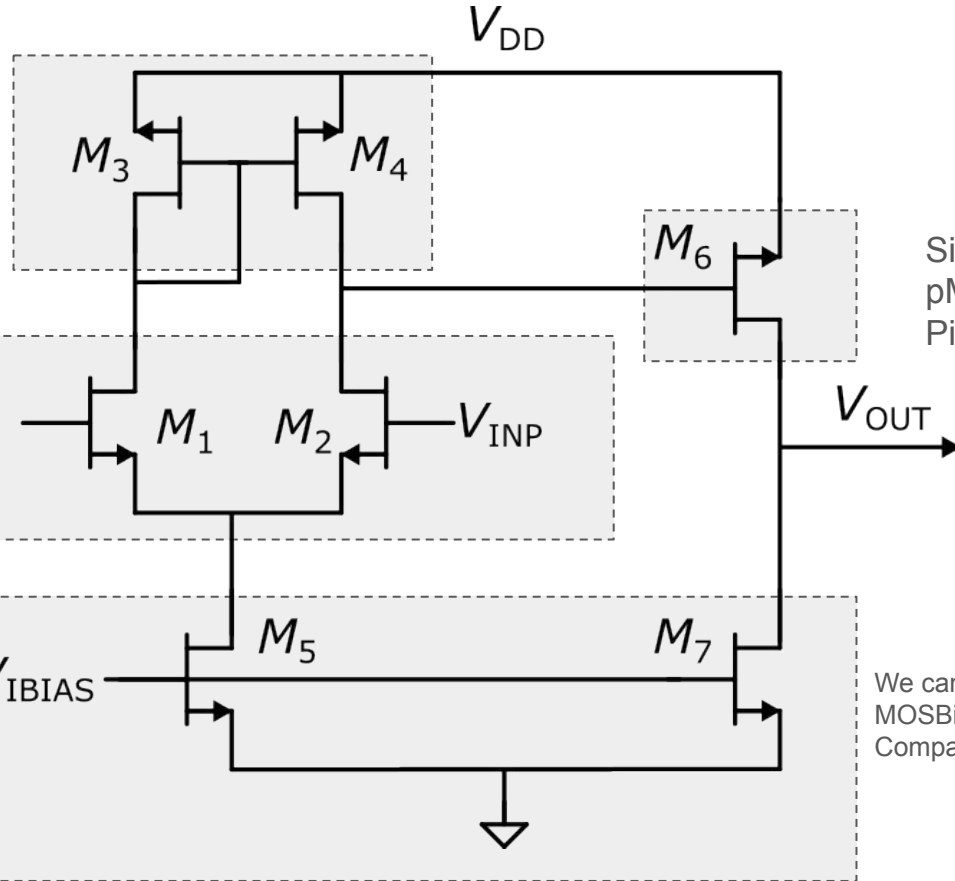
- Is an internal ladder for the DAC advisable?

Circuit block diagram



2 stage OTA based comparator circuit

pMOS Current mirror pairs (7 Pins)
Ratio = 1:1(2 pairs), 1:2, 1:4, 1:8



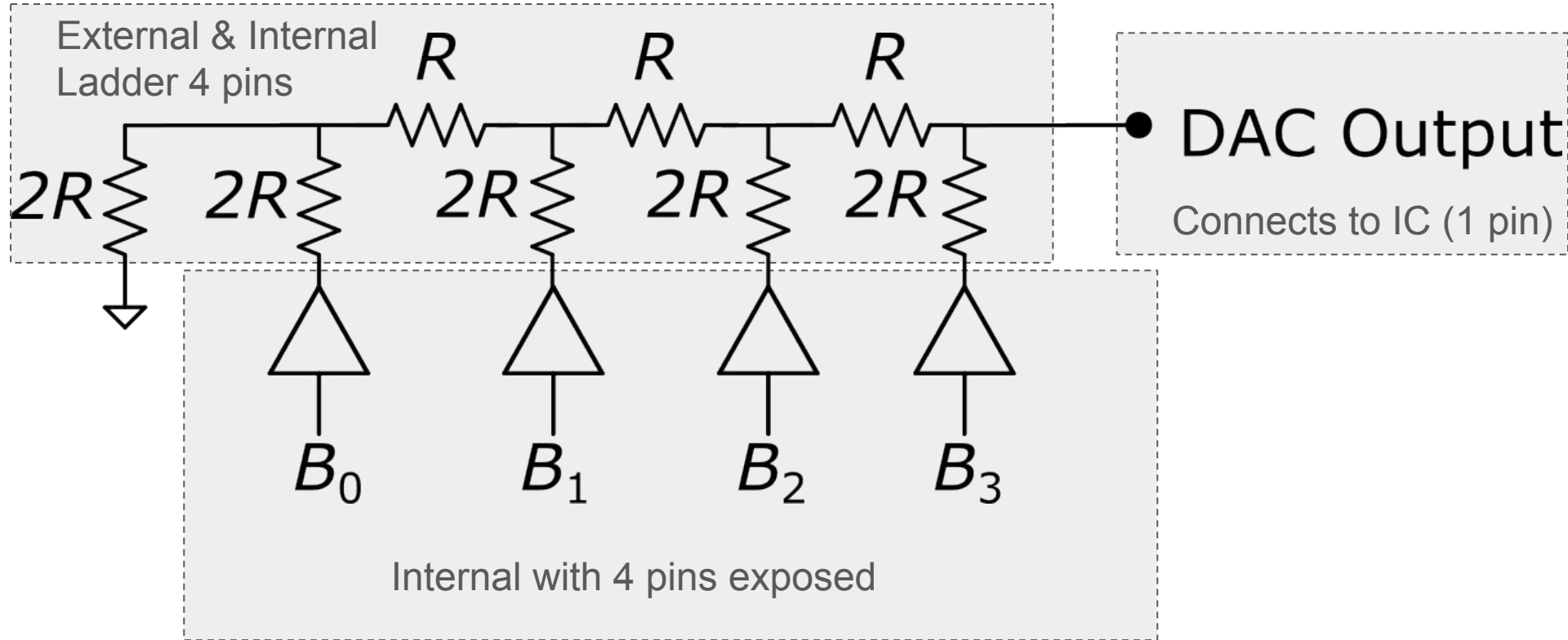
Size : 1x 4x
Differential nMOS
Pairs (13 Pins)

Size : 1x 4x
pMOS CS pair (4
Pins)

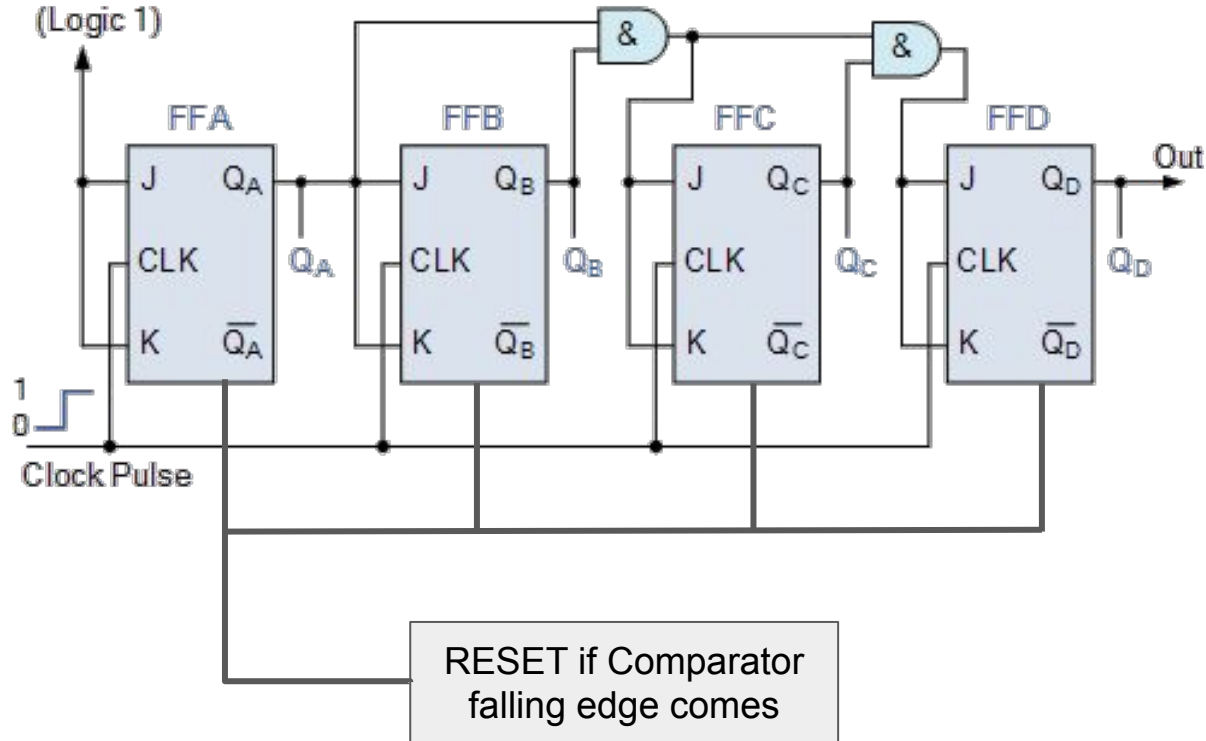
nMOS Current mirror pairs
(9 Pins)
Ratio = 1:1(2 pairs), 1:2, 1:4,
1:8, 1:16

We can leverage the existing
MOSBias as well for this
Comparator

R-2R DAC



4 Bit counter



This will be a fixed block and will not be reconfigurable or resizable through the switch matrix to keep the design simple

Pin map

